Qiskit Fall Fest 2026

Challenge 1: Predict the Output — Complete Solution

This solution implements the three required single-qubit circuits in Qiskit. Each circuit is measured for 100 shots and displayed as a histogram.

1. Circuit A — Apply one X gate

Prediction: The initial state is |0■. The Pauli-X gate flips |0■ to |1■, so an ideal circuit returns 1 in all 100 shots.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram
import matplotlib.pyplot as plt

simulator = AerSimulator()
qc_a = QuantumCircuit(1, 1)
qc_a.x(0)
qc_a.measure(0, 0)
print(qc_a.draw())
result_a = simulator.run(transpile(qc_a, simulator), shots=100).result()
counts_a = result_a.get_counts()
print(counts_a)  # Expected: {'1': 100}
plot_histogram(counts_a)
plt.show()
```

Expected ideal counts: {'1': 100}.

2. Circuit B — Apply two X gates

Prediction: The first X gate maps |0■ to |1■; the second maps |1■ back to |0■. Thus $X^2 = I$, and an ideal circuit returns 0 in all 100 shots.

```
qc_b = QuantumCircuit(1, 1)
qc_b.x(0)
qc_b.x(0)
qc_b.measure(0, 0)
print(qc_b.draw())
result_b = simulator.run(transpile(qc_b, simulator), shots=100).result()
counts_b = result_b.get_counts()
print(counts_b)  # Expected: {'0': 100}
plot_histogram(counts_b)
plt.show()
```

Expected ideal counts: {'0': 100}.

3. Circuit C — Apply one H gate

Prediction: H|0■ = (|0■ + |1■)/√2. The probabilities are P(0) = 0.5 and P(1) = 0.5. Over 100 shots, results are usually close to 50/50 but need not be exactly equal.

```
qc_c = QuantumCircuit(1, 1)
qc_c.h(0)
qc_c.measure(0, 0)
print(qc_c.draw())
result_c = simulator.run(transpile(qc_c, simulator), shots=100).result()
counts_c = result_c.get_counts()
print(counts_c)  # Example only: {'0': 51, '1': 49}
plot_histogram(counts_c)
plt.show()
```

Example counts such as {'0': 51, '1': 49} are illustrative, not guaranteed results. Run the code to obtain the actual simulator counts.

4. Combined comparison

Circuit	Gates	Prediction	Ideal / expected result
A	X	Always 1	100 × 1
B	X then X	Always 0	100 × 0
C	H	0 or 1	Approximately 50 × 0 and 50 × 1

5. Comparison and conclusion

Circuit A confirms that an X gate flips the computational-basis state. Circuit B shows that applying X twice returns the qubit to its original state. Circuit C creates an equal superposition; measurement can produce either 0 or 1, so it does not always produce the same result. Small differences from a 50/50 split are expected in finite-shot sampling.

Installation: If needed, run `%pip install qiskit qiskit-aer matplotlib` in a notebook cell, then restart the kernel.

Important: Qiskit was not installed in the preparation environment, so no live simulator results were generated here. The notebook contains runnable code that will generate the actual 100-shot counts and histograms.